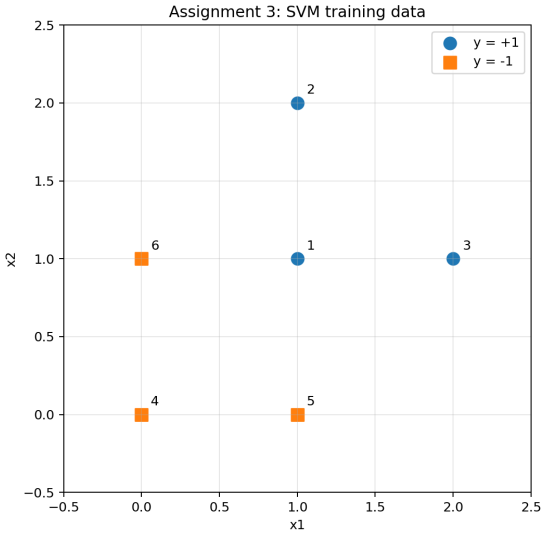
CSE 242 Assignment 3

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Question 1: Hard-margin SVM



(a) The support vectors are **points 1, 5, and 6**. Point 1 $(1, 1)$ is the positive point closest to the negatives, and points 5 $(1, 0)$ and 6 $(0, 1)$ are the negatives closest to it. Points 2, 3, and 4 are farther from the gap, so they don't affect the boundary.

(b) Support vectors satisfy $\mathbf{w} \cdot \mathbf{x}_i + b = y_i$:

$$\begin{aligned} \text{Point 1:} \quad & w_1 + w_2 + b = 1 \\ \text{Point 5:} \quad & w_1 + b = -1 \\ \text{Point 6:} \quad & w_2 + b = -1 \end{aligned}$$

Subtracting the second and third equations from the first gives $w_2 = 2$ and $w_1 = 2$, so $b = -3$. The boundary is

$$2x_1 + 2x_2 - 3 = 0.$$

All other points satisfy $y_i(\mathbf{w} \cdot \mathbf{x}_i + b) = 3 \geq 1$, so the constraints hold.

The geometric margin (distance from the boundary to the closest points) is

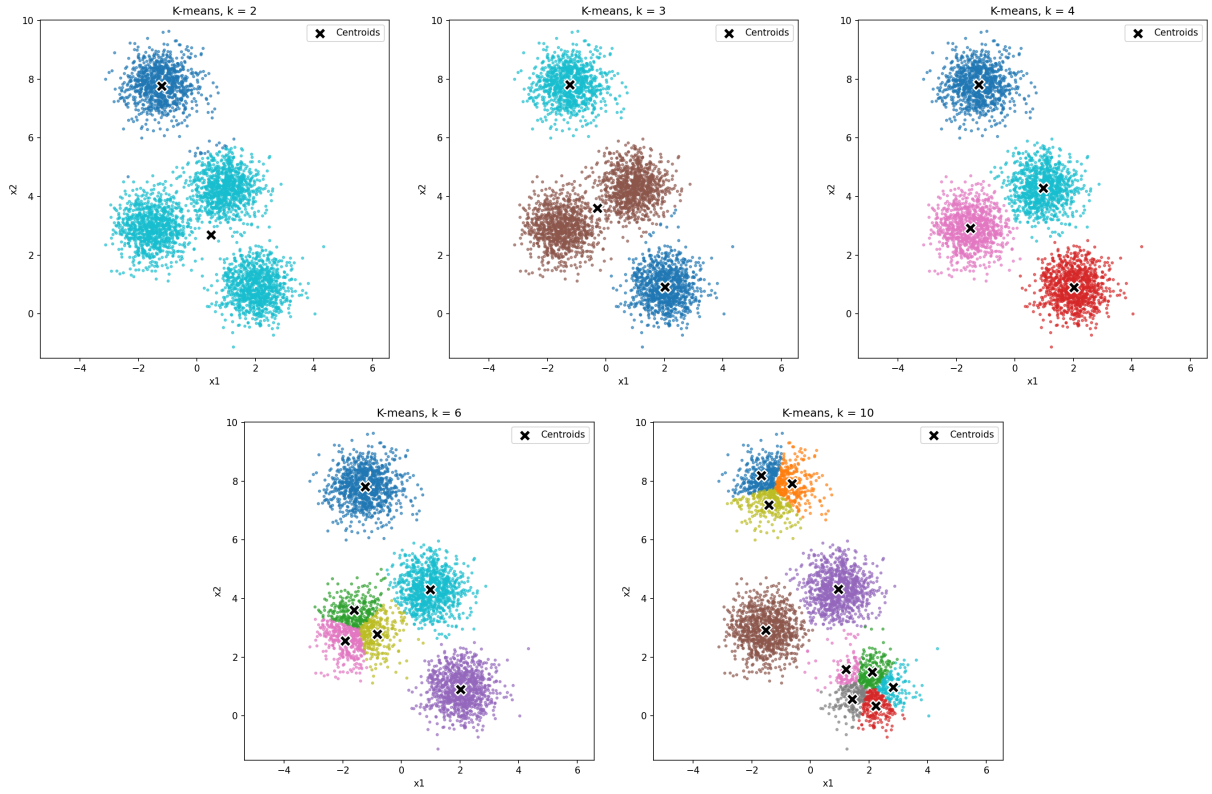
$$\frac{1}{\|\mathbf{w}\|} = \frac{1}{2\sqrt{2}} \approx 0.354.$$

The full width between the supporting lines is $2/\|\mathbf{w}\| \approx 0.707$.

Question 2: K-means

(a) Each point is assigned to the nearest centroid by squared Euclidean distance; ties go to the lowest centroid index. Each centroid is then updated to the mean of its points. An empty cluster keeps its previous centroid.

(b)



With $k = 2$ and $k = 3$, there are fewer clusters than blobs, so nearby blobs get merged. At $k = 4$, the clusters match the four generating blobs almost exactly. With $k = 6$ and $k = 10$, K-means starts splitting real blobs into pieces, and where those splits happen depends on the initial centroids rather than any real structure in the data.

Question 3: Noisy labels and K-NN repair

Over 20 trials (seed 0):

Method	Mean test accuracy
Baseline (noisy labels)	0.8280
K-NN repair	0.8644
Difference	+0.0364

K-NN repair helped, improving accuracy in all 20 trials. Because positive labels are flipped more often than negative ones, the baseline SVM is biased toward predicting negative. Majority voting among neighbors fixes many of the scattered wrong labels, which removes most of that bias. It could hurt near the class boundary, where neighbors are mixed and correct labels can

get overwritten, or if the noise were clustered so a whole neighborhood shared the wrong label. Per-trial results are in `noise_comparison.csv`.

Acknowledgment

I used Claude Code for debugging, and to write a test suite to check my answers. I also had it help with some of the latex formatting for this document.